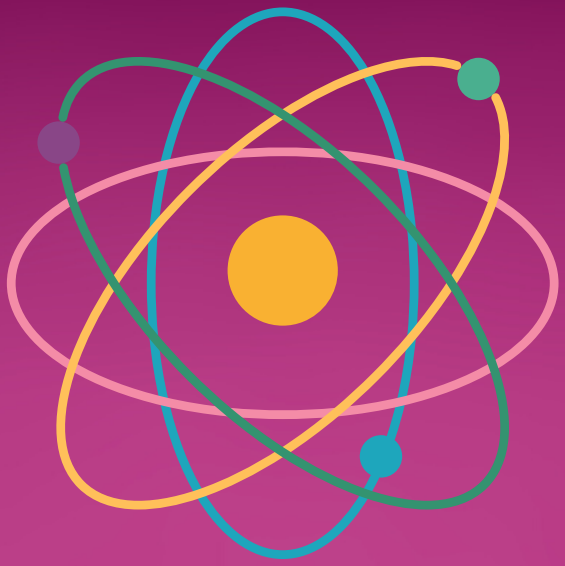
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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Solid State

English Medium

EXERCISE : S-I

CUBIC CRYSTALS

1. Number of crystal systems having, only 2 types of Bravais lattices = x,
Number of crystal systems having, atleast 2 interfacial angles equal = y, no. of crystal systems having all the three interfacial angles and all the three axes lengths equal = z
Then find y – (x + z).

SS0010

2. A closed packed structure of uniform spheres has the edge length of 600 pm. Calculate the radius of sphere in pm, if it exist in $(\sqrt{2}=1.4, \sqrt{3}=1.7)$

(a) simple cubic lattice (b) BCC lattice (c) FCC lattice

SS0001

3. Xenon crystallizes in the face-centred cubic lattice and the edge length of the unit cell is $438\sqrt{2}$ pm. What is the nearest neighbour distance in pm and what is the radius of xenon atom in pm?

SS0002

4. An element 'X' crystallizes in bcc. Find volume of unit cell in $(Å)^3$, if atomic radius is $\sqrt{3}$ Å.

SS0007

5. An element X (molar mass = 80 g/mol) having fcc structure, If number of unit cells in 8 gm of X are $N \times 10^{23}$ then value of N will be :

SS0163

6. If a lattice has simple cubic unit cell then number of faces which meet at a corner of a unit cell in this lattice is ?

SS0008

7. An element crystallizes in a face centered cubic lattice crystal and the edge of the unit cell is 0.5 nm. The density is 3.19 g/cm^3. What is the atomic weight of element (in $gmol^{-1}$) ?
[Give your answer as nearest integer]

SS0164

8. How many next nearest neighbours does each potassium atom have in its bcc lattice ?

SS0009

9. If number of nearest neighbours, next nearest (2nd nearest) neighbour and next to next nearest (3rd nearest) neighbours are x, y and z respectively for body centred cubic unit cell, then calculate value of $\frac{xy}{z}$? [Give your answer as nearest integer]

SS0012

10. A cubic solid is made up of two elements 'A' and 'B'. Atoms 'B' are at the corners of the cube and 'A' at the body centre. If empirical formula of compound is A_xB then value of x will be.

SS0006

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11. The effective radius of the iron atom is $\sqrt{2}$ Å. Iron has FCC structure. Calculate its density in $\frac{gm}{cm^3}$ (Fe = 56 amu , $N_A = 6 \times 10^{23}$) .

SS0165

12. Potassium has body-centred cubic structure with the nearest neighbour distance $260\sqrt{3}$ pm . Its density in gm/cm^3 would be ($\frac{1}{(5.2)^3} = 0.007$, $N_A = 6 \times 10^{23}$, K = 39)

SS0005

13. An element crystallizes in a structure having FCC unit cell of an edge 200 pm. Calculate the density in gm/cm^3, if 200 g of this element contains 2.4×10^{24} atoms. ($N_A = 6 \times 10^{23}$)

SS0019

14. The ratio of densities if same element which can exist as different allotropes in fcc as well as simple cubic packing is x. So value of x will be ? (Assume same atomic radius in both crystals)

SS0166

PACKING IN SOLIDS

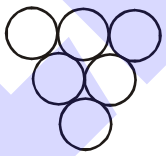
15. The density of solid Argon is 1.6 gm/ml at –233°C. If the atomic volume of Argon is assumed to be $\frac{5}{3}\times10^{-23}$ cm^3 then what % of solid Argon is apparently empty space ? (Ar = 40, $N_A = 6 \times 10^{23}$)

SS0018

16. In FCC unit cell, what fraction of edge is not covered by atoms ?

SS0013

17. A 3D unit cell is such that one of its planes has the following arrangement of atoms.



What will be the number of next nearest neighbour in such unit cell?

SS0017

18. Find packing fraction of three dimensional unit cell of AAAAA.....type hypothetical arrangement of identical atoms in which hexagonal packing arrangement is present in each layer ?

SS0020

19. Element A crystallizes in hcp. If total number of tetrahedral voids in 24 micrograms of A are $N \times 10^{17}$ then N will be ? ($N_A = 6 \times 10^{23}$, Atomic mass of A = 48)

SS0014

20. A compound forms hexagonal close-packed structure. What is the total mole of voids in 0.5 mole of this compound ?

SS0167

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21. An element 'M' crystallizes in ABAB....type packing. If adjacent layer A & B are $10\frac{\sqrt{2}}{\sqrt{3}}$ pm apart, then calculate radius of largest sphere in pm. which can be fitted in the void without disturbing the lattice arrangement (Given : $\sqrt{2} = 1.414$)

SS0016

22. For ABC ABC ABC .... packing, distance between two successive tetrahedral void is X and distance between two successive octahedral void is y in an unit cell, then $\frac{y\sqrt{2}}{X}$ is

SS0015

23. An element (atomic weight = 125) crystallises in simple cubic structure. Diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. If the density of element 'x' gm/cm^3 , the value of $\left(\frac{6x}{5}\right)$ is- [Given : $\sqrt{3} = 1.732$, $N_A = 6 \times 10^{23}$] **SS0168**

24. A cubic solid is made by atoms 'A' forming close packed arrangement. If Atoms 'B' occupy 0.25 fraction of tetrahedral voids then ratio of number of A to number of B atoms in solid is ?
[Give your answer as nearest integer] **SS0011**

25. A solid crystallises in close packing for 'P' atoms and 25% of tetrahedral voids are occupied by 'Q' atoms. If oxidation state of P is –2 then oxidation state of Q will be ?

SS0169

26. In compound M_2O_3, if oxide ions are in close packing while 'M^{3+}' cations occupy some of the voids. Then what fraction of total voids (O.V. & T.V.) are left unoccupied -

SS0170

IONIC CRYSTALS

27. The two ions A^+ and B^- have radii 88 pm and 200 pm respectively. In the closed packed crystal of compound AB, predict the co-ordination number of A^+.

SS0171

28. If the radius of Mg^{2+} ion, Cs^+ ion, O^{2-} ion, S^{2-} ion and Cl^- ion are 0.65 Å , 1.69 Å, 1.40 Å, 1.84 Å, and 1.81 Å respectively, calculate the co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl. **SS0021**

29. Spinel is a important class of oxides consisting of two types of metal ions with the oxide ions arranged in CCP pattern. The normal spinel has one-eight of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral hole occupied by another type of metal ion. Such a spinel is formed by Zn^{2+}, Al^{3+} and O^{2-}, with Zn^{2+} in the tetrahedral holes. If formula of spinel is $Zn_xAl_yO_4$ then $\frac{x}{y}$ will be.

SS0023

30. KF crystallizes in the NaCl type structure. If the radius of K^+ ion is 132 pm and that of F^- ion is 135 pm, what is the shortest K–F distance? What is the edge length of the unit cell? What is the closet $K^+ - K^+$ distance? [calculate all distances in pm]

SS0024

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31. CdO has NaCl structures with density 8.27 gcm^{-3}. If the ionic radius of O^{2-} is 1.24 Å, determine ionic radius of Cd^{2+} in Å (moler mass of CdO = 128) [$(103.184)^{1/3} = 4.69$]

SS0172

32. CsCl has bcc unit cell with edge length 400 pm. Calculate the interionic distance in CsCl in pm

SS0025

33. The density of KBr is 2.38 g cm^{-3} . The length of the edge of the unit cell is 700 pm. Find the number of fomula unit of KBr present in the single unit cell.

($N_A = 6 \times 10^{23}$ mol^{-1} , At. mass : K = 39, Br = 80)

[Give your answer as nearest integer]

SS0026

34. A crystal of lead(II) sulphide has NaCl structure. In this crystal the shortest distance between Pb^{+2} ion and S^{2-} ion is 300 pm. What is the length of the edge in pm of the unit cell in lead sulphide?

SS0027

[Give your answer as nearest integer]

35. If the length of the body diagonal for CsCl which crystallises into a cubic structure with Cl^- ions at the corners and Cs^+ ions at the centre of the unit cell, is 7 Å and the radius of the Cs^+ ion is 1.69 Å, what is the radius of Cl^- ion in Å?

SS0028

36. RbI crystallizes in (8 : 8) structure in which each Rb^+ is surrounded by eight iodide ions each of radius 2.17 Å. Find the length of one side of RbI unit cell in Å assuming anion, anion contact.

SS0029

37. Solid AB has NaCl type structure. If the radius of A^+ and B^- are 0.8 Å and 1.2Å respectively and formula mass of AB is 48 g/mole, what is the density of AB solid in gm/cm^3 ?

Take : Avogadro's number = 6×10^{23}

[Give your answer as nearest integer] **SS0030**

38. Calculate C–C bond length in (pm) in unit cell of diamond structure with edge length 698 pm.
[Diamond has ZnS type structure with carbon atoms occupying locations of Zn^{+2} & S^{-2} ions.]

SS0173

PROBLEMS RELATED WITH DEFECTS IN SOLID

39. The pyknometric density of sodium chloride crystal is 2.165×10^3 kg m^{-3} while its X–ray density is 2.178×10^3 kg m^{-3}. The percentage of unoccupied sites in sodium chloride crystal is :

[Hint : Percentage of unoccupied sites $= \frac{\text{X – ray density – pyknometric density}}{\text{X – ray density}} \times 100$]

SS0174

40. AgCl has the same structure as that of NaCl. The edge length of unit cell of AgCl is found to be 523.5 pm and the density of AgCl is 6.0 g cm^{-3}. Find the percentage of sites that are unoccupied.

[Ag = 108 , $(5.235)^3 = 143.5$] **SS0043**

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41. The density of ZnS crystal (Zinc blende structure) in gm/ml having 10% Frenkel defect is

[$r_{Zn^{2+}} = 40\sqrt{3}$pm, $r_{S^{2-}} = 110\sqrt{3}$pm , Zn = 65.2, S = 32] [$6^4 = 1296$] **SS0035**

[Give your answer as nearest integer]

42. If NaCl is dopped with 10^{-3} mol % $SrCl_2$, If number of cation vacancies per mole of NaCl are?

$N \times 10^{18}$ then value of N will be ? ($N_A = 6 \times 10^{23}$)

SS0032

43. The composition of a sample of wustite is $Fe_{0.93}O_{1.0}$. What percentage of iron is present in the form of Fe(III)?

SS0031

44. A certain sample of cuprous sulphide is found to have composition $Cu_{1.8}S$, because of presence of some Cu^{2+} ions in the lattice. Find mole % of copper which exists in form of cupric ions ?

[Give your answer as nearest integer]

SS0175

45. A non stoichiometric compound Fe_7S_8 consist of iron in both Fe^{+2} and Fe^{+3} forms and sulphur is present as sulphide ions. Calculate cation vacancies as a mole percentage of Fe^{+2} initially present in the ideal crystal ?

SS0034

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EXERCISE : S-II

1. An element (atomic weight = 125) crystallises in simple cubic structure. Diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. If the density of element 'x' gm/cm^3 , the value of $\left(\frac{6x}{5}\right)$ is -

[**Given** : $\sqrt{3}$ = 1.732 , $N_A = 6 \times 10^{23}$]

SS0036

2. The density of diamond from the fact that it has face centred cubic structure with two atoms per lattice point and unit cell edge length of 3.6 Å, is 'x' gm / cm^3, then the value of (1.458x) is ($N_A = 6 \times 10^{23}$)

SS0037

3. Iron crystallizes in several modifications. At about 910°C, the body-centred cubic 'α' form undergoes a transition to the face-centred cubic 'γ' form. Assuming that the distance between nearest neighbours is the same in the two forms at the transition temperature, the ratio of the density of γ iron to that of α iron at the transition temperature, x : 1, then the value of $(3\sqrt{6x})$ is.

SS0038

4. What is the percent by mass of titanium in rutile, a mineral that contain titanium and oxygen, if structure can be described as a closest packed array of oxide ions, with titanium in one half of the octahedral holes. What is the oxidation number of titanium? (Ti = 48)

SS0039

5 . An element crystallizes into a structure which may be described by a cubic type of unit cell having one atom on each corner of the cube and two extra atoms on one of its body diagonal. If the volume of this unit cell is 2.4×10^{-23} cm^3 and density of element is 7.2 g cm^{-3}, calculate the number of atoms present in 288 g of element.

SS0040

6. What will be packing fraction of solid in which atoms are present at corners and cubic void is occupied. The insertion of the sphere into void does not disturb simple cubic lattice.

[Given : $(\sqrt{3}-1)^3 = 0.39$]

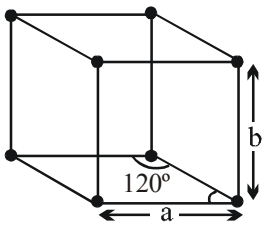
SS0041

7. Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined,

the lattice constants where **a** = 5 Å, and **b** = $3.5\sqrt{3}$ Å

How many molecules are contained in the given unit cell?

[density (ice) = $\frac{16}{17.5}$ gm/cm^3)]



SS0042

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8. The mineral hawleyite, one form of CdS, crystallizes in one of the cubic lattices, with edge length 5.87Å. The density of hawleyite is 4.63 g cm^{-3}. (Cd = 112)
(i) In which cubic lattice does hawleyite crystallize?
(ii) Find the Schottky defect in g cm^{-3}.

SS0043

9. KCl crystallizes in the same type of lattice as does NaCl. Given that $\frac{r_{Na^+}}{r_{Cl^-}}=0.5$ and $\frac{r_{Na^+}}{r_{K^+}}=0.7$
Calculate:

(a) The ratio of the sides of unit cell for KCl to that for NaCl and

(b) The ratio of densities of NaCl to that for KCl.

$\left(\left(\frac{8}{7}\right)^3=1.5, \frac{74.5}{58.5}=1.25\right)$

SS0044

10. A cubic unit cell contains manganese ions at the corners and fluoride ions at the center of each edge.

(a) What is the empirical formula of the compound?

(b) What is the co-ordination number of the Mn ion?

(c) Calculate the edge length of the unit cell, if the radius of Mn ion is 0.65 Å and that of F^- ion is 1.36 Å.

SS0045

11. Potassium crystallizes in a body-centred cubic lattice with edge length, a = 5.2 Å.
(a) What is the distance between nearest neighbours?
(b) What is the distance between next-nearest neighbours?
(c) How many nearest neighbours does each K atom have?
(d) How many next-nearest neighbours does each K atom have?
(e) What is the density of crystalline potassium?
(Given : K = 39, $(5.2)^3 = 140$)

SS0046

12 An element X (atomic weight = 24 gm/mole) forms a face centred cubic lattice. If the edge length of the lattice is 4×10^{-8} cm and the observed density is 2.40×10^3 kg/m^3 , calculate the percentage occupancy of lattice points by X element. (Given : $N_A = 6 \times 10^{23}$)

SS0047

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13 Calculate the density (in gm/cm^3) of NaCl type ionic solid (Molar mass = 75 gm/mol) if distance between two nearest cations is $250\sqrt{2}$ pm (avogadro number = 6×10^{23})

SS0048

14. The olivine series of minerals consists of crystals in which Fe and Mg ions may substitute for each other causing substitutional impurity defect without changing the volume of the unit cell. In olivine series of minerals, oxide ion exist as FCC with Si^{4+} occupying 1/4 th of octahedral voids and divalent ions occupying 1/4th of tetrahedral voids. The density of forsterite (magnesium silicate) is 3.0 g/cc and that of fayalite (ferrous silicate) is 4.0 g/cc. Find the formula of forsterite and fayalite minerals and the mass percentage of fayalite in an olivine with a density of 3.5 g/cc.

SS0049

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EXERCISE : O-I

***SINGLE CORRECT* :**

1. Which of the following are the correct axial distances and axial angles for rhombohedral system ?

(A) $a = b = c$, $\alpha = \beta = \gamma \neq 90^0$ (B) $a = b \neq c$, $\alpha = \beta = \gamma = 90^0$

(C) $a \neq b = c$, $\alpha = \beta = \gamma = 90^0$ (D) $a \neq b \neq c$, $\alpha \neq \beta \neq \gamma \neq 90^0$

SS0050

2. $a \neq b \neq c$, $\alpha \neq \beta \neq \gamma \neq 90^0$ represents

(A) tetragonal system (B) orthorhombic system

(C) monoclinic system (D) triclinic system

SS0051

3. Diamond belongs to the crystal system :

(A) Cubic (B) triclinic (C) tetragonal (D) hexagonal

SS0052

4. A match box exhibits -

(A) Cubic geometry (B) Monoclinic geometry

(C) Tetragonal geometry (D) Orthorhombic geometry

SS0053

5. Which of the following solids substances will have same refractive index when measured in different directions?

(A) NaCl (B) Monoclinic sulphur (C) Rubber (D) Graphite

SS0054

6. Crystal system in which maximum number of Bravais lattices are possible is

(A) Cubic (B) Triclinic (C) Orthorhombic (D) Rhombohedral

SS0056

7. In the body-centred cubic unit cell & face centred cubic unit cell, the radius of atom in terms of edge lengh(a) of the unit cell is respectively:

(A) $\frac{a}{2}, \frac{a}{2\sqrt{2}}$ (B) $\frac{a}{2\sqrt{2}}, \frac{\sqrt{3}a}{4}$ (C) $\frac{\sqrt{3}a}{4}, \frac{a}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}a}{2}, \frac{a}{2\sqrt{2}}$

SS0055

8. In which of the following arrangement distance between two nearest neighbours is maximum, considering identical sized atoms in all arrangements ?

(A) Simple cubic (B) bcc (C) fcc (D) equal in all

SS0069

9. Number of atoms at second nearest position from a given atom in a BCC structure is

(A) 8 (B) 6 (C) 12 (D) 4

SS0057

10. Correct sequence of the coordination number in SC, FCC & BCC is-

(A) 6, 8, 12 (B) 6, 12, 8 (C) 8, 12, 6 (D) 8, 6, 12

SS0059

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11. In a face centred cubic arrangement of A and B atoms, atoms of A are at the corner of the unit cell and atoms of B are at the face centres. One of the A atom is missing from one corner in unit cell. The simplest formula of compound is :-

(A) A_7B_3 (B) AB_3 (C) A_7B_{24} (D) $A_{7/8}B_3$

SS0176

12. Packing fraction in 2-D hexagonal arrangement -

(A) $\frac{\pi}{3\sqrt{2}}$ (B) $\frac{\pi}{3\sqrt{3}}$ (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\pi/6$

SS0066

13. Percentage area of each face covered by atoms in a FCC unit cell is -

(A) 60.4% (B) 68% (C) 74% (D) 78.5%

SS0058

14. The fraction of total volume ocupied by the atoms present in FCC is :-

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3\sqrt{2}}$ (C) $\frac{\pi}{4\sqrt{2}}$ (D) $\frac{\pi}{4}$

SS0177

15. Which one of the following schemes of ordering closest packed sheets of equal sized spheres do not generate close packed lattice.

(A) ABCABC (B) ABACABAC

(C) ABBAABBA (D) ABCBCABCBC

SS0064

16. What is not true regarding hexagonal close packing (hcp)

(A) packing fraction is 0.74

(B) coordination number is 12

(C) ABC ABC.....type packing

(D) Containing both tetrahedral and octahedral voids

SS0068

17. The interstitial hole is called tetrahedral because

(A) It is formed by four spheres.

(B) Partly same and partly different.

(C) It is formed by four spheres the centres of which form a regular tetrahedron.

(D) None of the above three.

SS0061

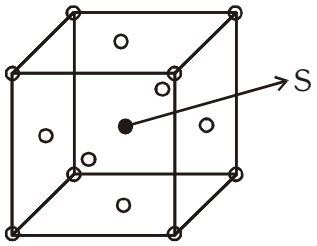
18. The size of an octahedral void formed in a closest packed lattice as compared to tetrahedral void is

(A) Equal (B) Smaller (C) Larger (D) Not definite

SS0062

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19. The site S in the adjoining fcc structure represents



(A) Octahedral void (B) Tetrahedral void (C) Triangular void (D) Square void

SS0178

20. If 'Z' is the number of atoms in the unit cell that represents the closest packing sequence ---A B C A B C ---, the number of tetrahedral voids in the unit cell is equal to

(A) Z (B) 2Z (C) Z/2 (D) Z/4

SS0060

21. In fcc unit cell smallest distance between octahedral void & tetrahedral void is -
(a = edge length of unit cell)

(A) $\frac{a}{\sqrt{2}}$ (B) $\frac{\sqrt{3}a}{2}$ (C) a (D) $\frac{\sqrt{3}a}{4}$

SS0067

22 . Copper metal crystallizes in FCC lattice. Edge length of unit cell is 362 pm. The radius of largest atom that can fit into the voids of copper lattice without disturbing it is

(A) 53 pm (B) 45 pm (C) 93 pm (D) 87 pm

SS0065

23. If the anions (A) form hexagonal closest packing and cations (C) occupy only 2/3 octahedral voids in it, then the general formula of the compound is

(A) CA (B) CA_2 (C) C_2A_3 (D) C_3A_2

SS0063

24. In a solid, oxide ions are arranged in ccp, cations A occupy $\left(\frac{1}{8}\right)^{th}$ of tetrahedral voids and cations B occupy $\left(\frac{1}{4}\right)^{th}$ of octahedral voids. The formula of compound is :-

(A) ABO_4 (B) AB_2O_3 (C) A_2BO_4 (D) AB_4O_4

SS0179

25. Atoms of element X form hcp lattice and atoms of element Y occupy "1/n th" of tetradedral voids. If the formula of compound formed by elements X and Y is X_2Y_3, then find "n".

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{1}{4}$

SS0180

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26. How many unit cell are there in 1 gram cubic crystal of NaCl ?

(A) $\frac{4\times N_A}{58.5}$ (B) $\frac{N_A}{58.5}$ (C) $\frac{N_A}{58.5\times 4}$ (D) $\frac{N_A}{58.5\times 8}$

SS0070

27. Edge length of M^+X^- (NaCl structure) is 7.2 Å. Assuming $M^+ - X^-$ contact along the cell edge, radius of X^- ion is ($r_{M^+} = 1.6Å$) :

(A) 2.0 Å (B) 5.6 Å (C) 2.8 Å (D) 38 Å

SS0076

28. $r_{Na^+} = 95$ pm and $r_{Cl^-} = 181$ pm in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions?

(A) 778.3 pm (B) 276 pm (C) 195.7 pm (D) 390.3 pm

SS0078

29. AB crystallises itself as NaCl crystal. If $r_+ = \frac{2}{\sqrt{6}}$ and $r_- = \sqrt{6}$, the edge length of cube is

(A) $2\sqrt{3}$ (B) $\frac{4}{\sqrt{3}}$ (C) $\frac{8}{\sqrt{6}}$ (D) $\frac{16}{\sqrt{6}}$ **SS0079**

30. A compound XY crystallizes in 8 : 8 lattice with unit cell edge lenght of 480 pm. If the radius of Y^- is 225 pm, then the radius of X^+ is

(A) 127.5 pm (B) 190.68 pm (C) 225 pm (D) 255 pm

SS0073

31. The mass of a unit cell of CsCl corresponds to

(A) 1 Cs^+ and 1 Cl^- (B) 1 Cs^+ and 6 Cl^- (C) 4 Cs^+ and 4 Cl^- (D) 8 Cs^+ and 1 Cl^-

SS0074

32. NH_4Cl crystallizes in CsCl type lattice with a unit cell edge length of 387 pm. The distance between the oppositively charged ions in the lattice is

(A) 335.1 pm (B) 83.77 pm (C) 274.46 pm (D) 137.23 pm

SS0077

33. An ionic compound AB has ZnS type structure. If the radius A^+ is 22.5 pm, then the ideal radius of B^- would be

(A) 54.35 pm (B) 100 pm (C) 145.16 pm (D) none of these

SS0075

34. The coordination number of cation and anion in Fluorite CaF_2 and CsCl are respectively

(A) 8:4 and 6:3 (B) 6:3 and 4:4 (C) 8:4 and 8:8 (D) 4:2 and 2:4

SS0072

35. The density of CaF_2 (fluorite structure) is 3.18 g/cm^3. The length of the side of the unit cell is (Ca = 40, F = 19)

(A) 253 pm (B) 344 pm (C) 546 pm (D) 273 pm

SS0071

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36. Radius of cation and anion are 2.5 Å & 2.6Å respectively. If a cubic crystal system is prepared by combination of above cation and anion then edge length of unit cell is (Take $\sqrt{3} = 1.7$)

(A) 2.7Å (B) 2Å (C) 3Å (D) 6Å

SS0181

37. Which of the following is the most likely to show schottky defect ?

(A) CaF_2 (B) ZnS (C) AgCl (D) CsCl **SS0080**

38. In the Schottky defect, in AB type ionic solids

(A) cations are missing from the lattice sites and occupy the interstitial sites

(B) equal number of cations and anions are missing

(C) anions are missing and electrons are present in their place

(D) equal number of extra cations and electrons are present in the interstitial sites **SS0081**

39. Choose the correct option.

(A) Two adjacent face centred atoms don't touch each other in fcc unit cell because they are not nearest atoms in lattice.

(B) Number of nearest Na^+ ions of another Na^+ in Na_2O crystal will be 24.

(C) Minimum distance between two cubical voids in simple cubic unit cell lattice will be 'a' where 'a' is length of edge of unit cell

(D) Due to defects in solids, density of solids either remains constant or decreases but it can never increase.

SS0082

40. The measured density of AgI is 6.94 g/cm^{-3} and the theoretical density is 5.67 g/cm^{-3}. These data indicate that solid AgI has -

(A) Schottky defect (B) Frenkel defect

(C) Interstitial impurities defect (D) Both (A) and (B)

SS0083

41. Which of the following statement is **CORRECT** ?

(A) A metal can show only non- stoichiometric defects

(B) Schottky defect reduces the density of a solid due to significant increase in volume.

(C) Impurity defect always change the density.

(D) Solids having F-centres may have metal excess defect due to missing anions. **SS0084**

42. Which of the following is/are correct statement (s).

(a) In frenkel defect, the larger ion gets dislocated from its normal site to an interstitial site.

(b) NaCl crystal contains greater number of schottky pairs per cm^3 solid as temperature increases.

(c) In crystal of ionic solids, the sites occupied by unpaired electrons are called F-centres.

(d) On heating zinc oxide, excess of Zn develops in the crystal and its formula becomes $Zn_{1+x}O$.

(A) a,b,c (B) a,b,d (C) b,c,d (D) c,d

SS0182

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43. Which of the following is NOT correct.

(A) Substances like $MgFe_2O_4$ lose their ferrimagnetism on heating and become paramagnetic

(B) Diamagnetic substances do not contain unpaired electrons.

(C) Magnetic moment associated with electron is measured in unit called Bohr Magneton.

(D) Domains of electrons in Anti-ferromagnetic substances are aligned in parallel and anti-parallel directions in unequal numbers.

SS0183

44. Alignment of magnetic moments in a substance is given as follows :-



The substance may be

(A) MnO (B) CrO_2 (C) Fe_3O_4 (D) NaCl

SS0184

45. A piece of Copper and another of Germanium are cooled from room temperature to 80 K. The resistance of:

(A) each of them increases

(B) Cu increases and that of Ge decreases

(C) Cu decreases and that of Ge increases

(D) each of them decreases

SS0185

ALLEN®

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EXERCISE : O-II

SINGLE CORRECT :

1. The only incorrect statement for the packing of identical spheres in two dimension is :

(A) For square close packing , coordination number is 4.

(B) For hexagonal close packing, coordination number is 6.

(C) There is only one void per atom in both, square and hexagonal close packing.

(D) Hexagonal close packing is more efficiently packed than square close packing. **SS0085**

2. Correct statement for ccp is :

(A) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 4 octahedral voids

(B) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 6 octahedral voids

(C) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 8 octahedral voids

(D) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 12 octahedral voids **SS0086**

3. Which of the following statements is correct in the rock-salt structure of an ionic compounds?

(A) coordination number of cation is four whereas that of anion is six.

(B) coordination number of cation is six whereas that of anion is four.

(C) coordination number of each cation and anion is four.

(D) coordination number of each cation and anion is six. **SS0087**

MORE THAN ONE MAY BE CORRECT :

4. Which of the following statements is/are correct :

(A) In an anti-fluorite structure, anions form FCC lattice and cations occupy all tetrahedral voids.

(B) If the radius of cations and anions are 0.2 Å and 0.95 Å, then coordination number of cation in the crystal is 4.

(C) Each sphere is surrounded by six voids in two dimensional hexagonal close packed layer.

(D) 8 Cs^+ ions occupy the second nearest neighbour locations of a Cs^+ ion in CsCl crystals.

5. Select correct statement(s) **SS0088**

(A) Density of crystal always increases due to substitutional impurity defect.

(B) An ion is transferred from a lattice site to an interstitial position in Frenkel defect.

(C) In AgCl, the silver ion is displaced from its lattice position to an interstitial position. Such a defect is called a frenkel defect

(D) None **SS0089**

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6. Lead metal has a density of 11.34 g/cm^3 and crystallizes in a face–centred lattice. Choose the correct alternatives (Pb = 208 , $N_A = 6 \times 10^{23}$)

(A) the volume of one unit cell is 1.22×10^{-22} cm^3.

(B) the volume of one unit cell is 1.22×10^{-19} cm^3.

(C) the atomic radius of lead is 175 pm.

(D) the atomic radius of lead is 155.1 pm. **SS0090**

7. Which of the following statement(s) is/are correct ?

(A) NaCl is a 'AB' crystal lattice that can be interpreted to be made up of two individual fcc unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.

(B) In a face centred cubic unit cell, the body centre is an octahedral void.

(C) In fcc unit cell, octahedral and tetrahedral voids are equal in number.

(D) Tetrahedral voids = 2 × octahedral voids, is valid for ccp and hcp. **SS0091**

8. Select the correct statement (s) :

(A) CsCl mainly shows Schottky defect

(B) ZnS mainly shows Frenkel defect

(C) NaCl unit cell contain $4Na^+$ and $4Cl^-$

(D) Truncated octahedron have 24 corners.

SS0092

9. Select the correct statement(s) –

(A) The ionic crystal of AgBr has Schottky defect.

(B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90°$, $\gamma = 120°$ is hexagonal

(C) Ionic compounds having Frenkel defect has high r^+/r^- ratio.

(D) The co-ordination number of Na^+ ion in NaCl is 6 **SS0093**

10. Which of the following is/are true ?

(A) Ratio of nearest neighbours in simple cubic cell to next nearest neighbours in face centred is cubic cell is 1.

(B) Packing efficiency of a unit cell in which atoms are present at each corner and each edge centre is about 26 % in metallic crystal.

(C) Distance between two planes in FCC or HCP arrangement is same for a metal existing in both forms, with same atomic radius.

(D) If number of unit cell along one edge are 'x' , then total number of unit cell in cube = x^3

SS0094

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ASSERTION / REASON :

11. **Statement-1** : In Antifluorite structure (Li_2O), the oxide ions occupy c.c.p. (cubic close packing) and Li^+ ions, 100% tetrahedral voids.

Statement-2 : The distance of the nearest neighbours in antifluorite structure is $\frac{\sqrt{3}a}{4}$, where 'a' is the edge length of the cube

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **SS0095**

12. **Statement-1 :** In FCC unit cell, packing efficiency is more when all tetrahedral voids are filled with spheres of maximum possible size as compared with packing efficiency when all octahedral voids are filled in similar way.

Statement-2 : Tetrahedral voids are more in the number than octahederal voids in FCC.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **SS0096**

13. **Statement-1 :** In diamond , carbon atoms occupy alternate tetrahedral voids in the FCC lattice formed by the carbon atoms.

Statement-2 : In diamond, packing fraction is more than 74%.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **SS0097**

14. **Statement-1 :** Due to Frenkel defect, there is no effect on the density of the crystalline solid.

Statement-2 : In Frenkel defect, no cation or anion leaves the crystal.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **SS0098**

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15. **Statement-1 :** Conductivity of silicon increased by doping it with group 15 element.
Statement-2 : Doping means introduction of small amount of impurities like P or As into pure silicon crytstal.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **SS0099**

COMPREHENSION :

Paragraph for Q.16 & Q.17

Solid balls of radius 17.32 cm crystallises in bcc pattern. During one such crystallisation, some oxygen gas is trapped. This trapped oxygen at 640K creates pressure of 5 atm.

Assume :

(i) BCC arrangement is not disturbed due to trapping of gas.

(ii) Gas is uniformly distributed inside unit cell

[Take R = 0.08 atm-litre/mole-K , $N_A = 6 \times 10^{23}$, Mass of a solid ball = 64 gms]

16. Number of oxygen molecules present in an unit cell is -

(A) 2.4×10^{24} (B) 1.2×10^{24} (C) 6×10^{23} (D) 3×10^{23} **SS0100**

17. Calculate percentage increase in density due to trapping of gas

(A) 16.67 % (B) 33.33 % (C) 100% (D) 50% **SS0101**

Paragraph for (Que. 18 to 21)

Calcium crystallizes in a cubic unit cell with density 3.2 g/cc. Edge-length of the unit cell is 437 pm.

18. The type of unit cell is

(A) Simple cubic (B) BCC (C) FCC (D) Edge-centred **SS0102**

19. The nearest neighbour distance is

(A) 154.5 pm (B) 309 pm (C) 218.5 pm (D) 260 pm **SS0103**

20. The number of nearest neighbours of a Ca atom are

(A) 4 (B) 6 (C) 8 (D) 12 **SS0104**

21. If the metal is melted, density of the molten metal was found to be 3 g/cc. What will be the percentage of empty space in the molten metal?

(A) 31% (B) 36% (C) 28% (D) 49% **SS0105**

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***TABLE TYPE* COMPREHENSION :**

	Column-I		Column-II		Column-III
(A)	NaCl (Rock salt) structure	(i)	Cation - FCC Anion - Tetrahedral void	(I)	All tetrahedral voids are occupied
(B)	CsCl structure	(ii)	Anion - FCC Cation - Tetrahedral voids	(II)	All octahedral voids are occupied
(C)	ZnS (zinc blende) structure	(iii)	Anion - SC Cation - Cubic voids	(III)	50 % of tetrahedral voids are occupied
(D)	CaF_2 (fluorite) structure	(iv)	Anion - FCC Cation - Octahedral void	(IV)	All octahedral voids are empty

22. Which of the following is correct match ?
(A) A, i, I (B) A, ii, IV (C) A, iv, II (D) A, iv, IV
SS0106

23. Which of the following is incorrect match ?
(A) B, iii, I (B) C, ii, III (C) D, i, I (D) D, i, IV
SS0107

24. Which of the following is correct match ?
(A) D, ii, I (B) B, iv, IV (C) D, iii, I (D) C, ii, IV
SS0108

MATCH THE COLUMN :

25. Match the column

	Column I		Column II
(A)	Tetragonal and Hexagonal	(P)	are two crystal systems
(B)	Cubic and Rhombohedral	(Q)	$a = b \neq c$
(C)	Monoclinic and Triclinic	(R)	$a \neq b \neq c$
(D)	Cubic and Hexagonal	(S)	$a = b = c$

SS0109

26. Match the column:

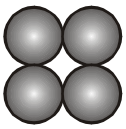
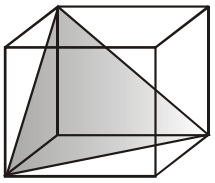
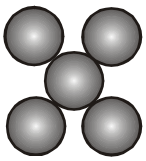
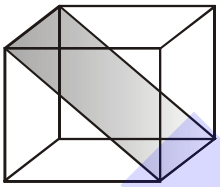
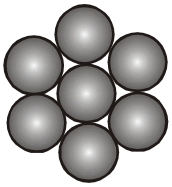
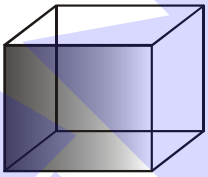
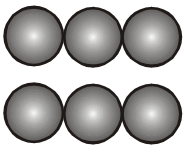
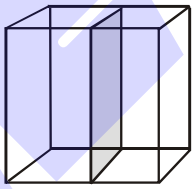
	Column I		Column II
(A)	Rock salt structure	(P)	Co-ordination number of cation is 4
(B)	Zinc Blend structure	(Q)	$\frac{\sqrt{3}a}{4} = r_+ + r_-$
(C)	Flourite structure	(R)	Co-ordination number of cation and anion are same
		(S)	Distance between two nearest anion is $\frac{a}{\sqrt{2}}$

SS0110

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***MATCHING LIST TYPE* :**

27. Match the column

Column I (Arrangement of the atoms/ions)		**Column II** (Planes in fcc lattice)	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	4	3	2	1
(C)	3	2	1	4
(D)	1	2	4	3

SS0111

28.

Column I [Distance in terms of edge length of cube (a)]		**Column II**	
(P)	0.866 a	(1)	Shortest distance between cation & anion in CsCl structure.
(Q)	0.707 a	(2)	Shortest distance between two cation in CaF_2 structure.
(R)	0.433 a	(3)	Shortest distance between carbon atoms in diamond.
(S)	a	(4)	shortest distance between next nearest cations in rock salt structrue.

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	1	2	3	4
(C)	3	2	1	4
(D)	1	2	4	3

SS0112

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Solid, Liquid & GK-(PC)\Eng\01- Solid State\02_Ex.p65

EXERCISE : J-MAIN

1. In a face centred cubic lattice, atoms of A form the corner points and atoms of B form the face centred points. If two atoms of A are missing from the corner points, the formula of the ionic compound is **[Jee-Main (online)-13]**

(1) AB_2 (2) AB_3 (3) AB_4 (4) A_2B_5 **SS0120**

2. Which one of the following statements about packing in solids is **incorrect** ?

(1) Void space in ccp mode of packing is 26% **[Jee-Main (online)-13]**

(2) Coordination number in hcp mode of packing is 12

(3) Void space in hcp mode of packing is 32%

(4) Coordination number in bcc mode of packing is 8 **SS0121**

3. An element having an atomic radius of 0.14 nm crystallizes in an fcc unit cell. What is the length of a side of the cell ? **[Jee-Main (online)-13]**

(1) 0.96 nm (2) 0.4 nm (3) 0.24 nm (4) 0.56 nm **SS0122**

4. Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :- **[Jee-Main (offline)-13]**

(1) 7.01% (2) 4.08% (3) 6.05% (4) 5.08 **SS0123**

5. The total number of octahedral void(s) per atom present in a cubic close packed structure is :- **[Jee-Main (online)-14]**

(1) 1 (2) 2 (3) 3 (4) 4 **SS0124**

6. In a monoclinic unit cell, the relation of sides and angles are respectively **[Jee-Main (online)-14]**

(1) $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$ (2) $a \neq b \neq c$ and $\beta = \gamma = 90^\circ \neq \alpha$

(3) $a = b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$ (4) $a \neq b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$ **SS0125**

7. The appearance of colour in solid alkali metal halides is generally due to : **[Jee-Main (online)-14]**

(1) Frenkel defect (2) F-centres (3) Schottky defect (4) Interstitial position **SS0126**

8 . In a face centred cubic lattice atoms A are at the corner points and atoms B at the face centred points. If atom B is missing from one of the face centred points, the formula of the ionic compound is : **[AIEEE-2011, Jee-Main (online)-14]**

(1) AB_2 (2) A_2B_3 (3) A_5B_2 (4) A_2B_5 **SS0127**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Solid, Liquid & CK-(PC)\Eng\01. Solid State\02_Ex.p65

9. CsCl crystallises in body centred cubic lattice. if 'a' is its edge length then which of the following expression is correct : **[Jee-Main (offline)-14]**

(1) $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}}{2}a$ (2) $r_{Cs^+} + r_{Cl^-} = \sqrt{3}a$ (3) $r_{Cs^+} + r_{Cl^-} = 3a$ (4) $r_{Cs^+} + r_{Cl^-} = \frac{3a}{2}$

SS0128

10. Sodium metal crystallizes in a body centred cubic lattice with a unit cell edge of 4.29Å. The radius of sodium atom is approximately :- **[Jee-Main (offline)-15]**

(1) 5.72Å (2) 0.93Å (3) 1.86Å (4) 3.022Å **SS0129**

11. Which of the following compounds is metallic and ferromagnetic ? **[Jee-Main (offline)-16]**

(1) MnO_2 (2) TiO_2 (3) CrO_2 (4) VO_2 **SS0130**

12. A metal crystallises in a face centred cubic structure. If the edge length of its unit cell is 'a', the closest approach between two atoms in metallic crystal will be :- **[Jee-Main (offline)-17]**

(1) 2a (2) $2\sqrt{2}\,a$ (3) $\sqrt{2}\,a$ (4) $\frac{a}{\sqrt{2}}$ **SS0131**

13. Which type of defect has the presence of cations in the interstitial sites -

(1) Vacancy defect (2) Frenkel defect **[Jee-Main (offline)-18]**

(3) Metal deficiency defect (4) Schottky defect **SS0132**

14. All of the following share the same crystal structure except :- **[Jee-Main (online)-18]**

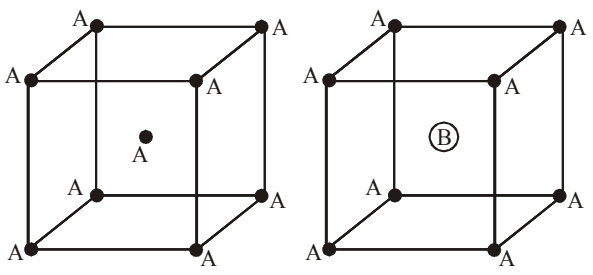
(1) RbCl (2) CsCl (3) LiCl (4) NaCl **SS0133**

15. Which of the following arrangements shows the schematic alignment of magnetic moments of antiferromagnetic substance ? **[Jee-Main (online)-18]**

(1) ↑ ↓ ↓ ↓ ↓ ↑ (2) ↑ ↑ ↑ ↑ ↑ ↑

(3) ↑ ↑ ↓ ↑ ↑ ↓ (4) ↑ ↓ ↑ ↓ ↑ ↓ **SS0134**

16. Consider the bcc unit cells of the solids 1 and 2 with the position of atoms as shown below. The radius of atom B is twice that of atom A. The unit cell edge length is 50% more in solid 2 than in 1. What is the approximate packing efficiency in solid 2? **[Jee-Main (online)Jan-19]**



Solid 1 Solid 2

(1) 45% (2) 65% (3) 90% (4) 75% **SS0113**

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17. The ratio of number of atoms present in a simple cubic, body centered cubic and face centered cubic structure are, respectively : **[Jee-Main (online)Jan-19]**

(1) 1 : 2 : 4 (2) 8 : 1 : 6 (3) 4 : 2 : 1 (4) 4 : 2 : 3

SS0114

18. An element has a face-centred cubic (fcc) structure with a cell edge of a. The distance between the centres of two nearest tetrahedral voids in the lattice is : **[Jee-Main (online)Jan-19]**

(1) $\frac{a}{2}$ (2) a (3) $\frac{3}{2}a$ (4) $\sqrt{2}\,a$

SS0115

19. At 100°C, copper (Cu) has FCC unit cell structure with cell edge length of x Å. What is the approximate density of Cu (in g cm^{-3}) at this temperature ? **[Jee-Main (online)April-19]**
[Atomic Mass of Cu = 63.55u]

(1) $\frac{105}{x^3}$ (2) $\frac{211}{x^3}$ (3) $\frac{205}{x^3}$ (4) $\frac{422}{x^3}$

SS0116

20. Which primitive unit cell has unequal edge lenghs ($a \neq b \neq c$) and all axial angles different from 90° **[Jee-Main (online)April-19]**

(1) Tetragonal (2) Hexagonal (3) Monoclinic (4) Triclinic

SS0116

21. The radius of the largest sphere which fits properly at the centre of the edge of body centred cubic unit cell is : (Edge length is represented by 'a') :- **[Jee-Main (online)April-19]**

(1) 0.134 a (2) 0.027 a (3) 0.067 a (4) 0.047 a

SS0117

22. A solid having density of 9×10^3 kg m^{-3} forms face centred cubic crystals of edge length $200\sqrt{2}$ pm. What is the molar mass of the solid ? **[Jee-Main (online)April-19]**
(Avogadro constant $\cong 6 \times 10^{23}$ mol^{-1}, $\pi \cong 3$)

(1) 0.0216 kg mol^{-1} (2) 0.0305 kg mol^{-1} (3) 0.4320 kg mol^{-1} (4) 0.0432 kg mol^{-1}

SS0118

23. Which of the following compounds is likely to show both Frenkel and Schottky defects in its crystalline form? **[Jee-Main (online)2020]**

(1) AgBr (2) ZnS (3) KBr (4) CsCl

SS0186

24. Ga (atomic mass 70 u) crystallizes in a hexagonal close packed structure. The total number of voids in 0.581 g of Ga is _____ $\times 10^{21}$. **[Round off to the Nearest Integer]. [Jee-Main (online)2021]**

SS0187

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Solid, Liquid & CK-(PC)\Eng\01. Solid State\02_Ex.p65

25. A certain element crystallises in a bcc lattice of unit cell edge length 27 Å. If the same element under the same conditions crystallises in the fcc lattice, the edge length of the unit cell in Å will be _____.
[Round off to the Nearest Integer]. **[Jee-Main (online)2021]**
[Assume each lattice point has a single atom]
[Assume $\sqrt{3} = 1.73$, $\sqrt{2} = 1.41$]

SS0188

26. The unit cell of copper corresponds to a face centered cube of edge length 3.596 Å with one copper atom at each lattice point. The calculated density of copper in kg/m^3 is _____.
[Molar mass of Cu : 63.54 g ; Avogadro Number = 6.022×10^{23}] **[Jee-Main (online)2021]**
[Round off to the Nearest Integer].

SS0189

27. The number of octahedral voids per lattice site in a lattice is _____.
[Round off to the Nearest Integer].

SS0190

28. Diamond has a three dimensional structure of C atoms formed by covalent bonds. The structure of diamond has face centred cubic lattice where 50% of the tetrahedral voids are also occupied by carbon atoms. The number of carbon atoms present per unit cell of diamond is _____.
[Round off to the Nearest Integer]. **[Jee-Main (online)2021]**

SS0191

29. A copper complex crystallising in a CCP lattice with a cell edge of 0.4518 nm has been revealed by employing X-ray diffraction studies. The density of a copper complex is found to be 7.62 g cm^{-3}.
The molar mass of copper complex is _____ g mol^{-1}. **[Jee-Main (online)2021]**
[Given : NA = 6.022×10^{23} mol^{-1}]
[Round off to the Nearest Integer].

SS0192

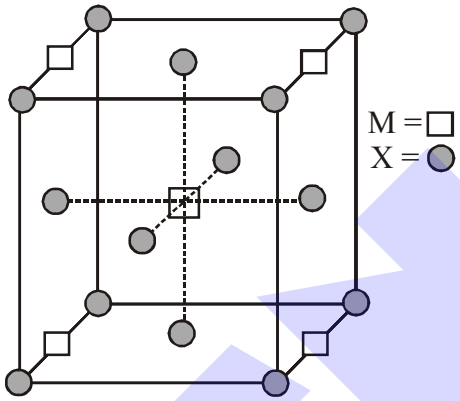
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Solid, Liquid & GK-(PC)\Eng\01-Solid State\02_Ex.p65

EXERCISE : J-ADVANCED

1. The number of hexagonal faces that present in a truncated octahedron is. **[JEE-2011]**

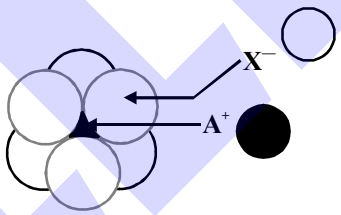
SS0135

2. A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is : **[JEE-2012]**



(A) MX (B) MX_2 (C) M_2X (D) M_5X_{14} **SS0136**

3. The arrangement of X^- ions around A^+ ion in solid AX is given in the figure (not drawn to scale). If the radius of X^- is 250 pm, the radius of A^+ is - **[JEE-2013]**



(A) 104 pm (B) 125 pm (C) 183 pm (D) 57 pm **SS0137**

4. If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions m and n respectively, are - **[JEE-2015]**

(A) $\frac{1}{2},\frac{1}{8}$ (B) $1,\frac{1}{4}$ (C) $\frac{1}{2},\frac{1}{2}$ (D) $\frac{1}{4},\frac{1}{8}$ **SS0138**

5. The **CORRECT** statement(s) for cubic close packed (ccp) three dimensional structure is (are)

(A) The number of the nearest neighbours of an atom present in the topmost layer is 12

(B) The efficiency of atom packing is 74% **[JEE-2016]**

(C) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively

(D) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

SS0139

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6. A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8g cm^{-3}, then the number of atoms present in 256g of the crystal is $N \times 10^{24}$. The value of N is **[JEE-2017]**

SS0140

7. Consider an ionic solid MX with NaCl structure. Construct a new structure (Z) whose unit cell is constructed from the unit cell of MX following the sequential instructions given below. Neglect the charge balance. **[JEE-2018]**

(i) Remove all the anions (X) except the central one

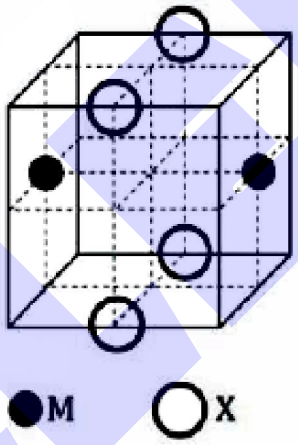
(ii) Replace all the face centered cations (M) by anions (X)

(iii) Remove all the corner cations (M)

(iv) Replace the central anion (X) with cation (M)

The value of $\left(\frac{\text{number of anions}}{\text{number of cations}}\right)$ in Z is____. **SS0141**

8. The cubic unit cell structure of a compound containing cation M and anion X is shown below. When compared to the anion, the cation has smaller ionic radius. Choose the correct statement(s). **[JEE-2020]**



(A) The empirical formula of the compound is MX.

(B) The cation M and anion X have different coordination geometries.

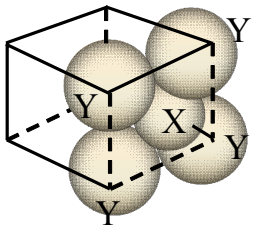
(C) The ratio of M-X bond length to the cubic unit cell edge length is 0.866.

(D) The ratio of the ionic radii of cation M to anion X is 0.414.

SS0193

9. For the given close packed structure of a salt made of cation X and anion Y shown below (ions of only one face are shown for clarity) , the packing fraction is approximately

(packing fraction = $\frac{\text{Packing efficiency}}{100}$) **[JEE-2021]**



(A) 0.74 (B) 0.63 (C) 0.52 (D) 0.48

SS0194

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ANSWER KEY

EXERCISE : S-I

1. 2.00
2. (a) 300.00 pm, (b) 255.00 pm, (c) 210.00 pm
3. 438.00 pm, 219.00 pm
4. 64.00
5. 0.15
6. 12.00
7. 60
8. 6.00
9. 4
10. 1.00
11. 5.83
12. 0.91
13. 41.67
14. 1.41
15. 60.00 %
16. 0.29
17. 6.00
18. 0.60
19. 6.00
20. 1.50
21. 2.07
22. 2.00
23. 2.00
24. 2
25. 4.00
26. 0.77 or 0.78
27. 6.00
28. 4.00, 6.00, 8.00
29. 0.50
30. 267.00, 534.00, 377.60
31. 1.10
32. 346.41
33. 4
34. 600
35. 1.81
36. 4.34
37. 5
38. 302.23
39. 0.59 – 0.60
40. 10.00
41. 3
42. 6.00
43. 15.05
44. 11
45. 12.50

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EXERCISE : S-II

1. 2

2. 5

3. 8

4. 60%, +4

5 . 5×10^{24}

6. 0.72 or 0.73

7. 4

8 (i) FCC (ii) 0.116 g/cc

9. (a) 1.14, (b) 1.20

10. (a) MnF_3, (b) 6, (c) 4.02Å

11. (a) 4.5 Å, (b) 5.2 Å, (c) 8, (d) 6, (e) 0.929 g/cm^3

12 96%

13. 4

14. Mg_2SiO_4, Fe_2SiO_4, 57.14%

EXERCISE : O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	A	D	C	C	C	D	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	D	B	C	C	C	C	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	C	A	C	C	A	D	D	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	A	B	C	C	D	D	B	C	C
Que.	41	42	43	44	45					
Ans.	D	C	D	A	C					

EXERCISE : O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	A,C	B,C	A,C	A,B,D	A,B,C,D	A,B,D	A,B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	C	A	A	B	D	C	B	D
Que.	21	22	23	24	25					
Ans.	A	C	A	D	A-P,Q	B-P,S	C-P,R	D-P		
Que.	26			27	28					
Ans.	A-R,S	B-P,Q,R,S	C-Q	A	B					

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EXERCISE : J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	2	2	1	2	2	4	1	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	2	2	4	3	1	1	4	4
Que.	21	22	23	24	25	26	27	28	29	
Ans.	3	2	1	15	33	9076	1	8	106	

EXERCISE : J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	
Ans.	8	B	A	A	B,C,D	2	3	A,C	B	

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Solid, Liquid & CK-(PC)\Eng\01 Solid State\02_Ex.p65